

Tilt EQ

A one-knob tilt equaliser with high- and low-pass filters and mid / side tilt.



Overview

Tilt EQ is in the spirit of the classic one-knob hi-fi and mastering tilt controls. One knob seesaws the whole spectrum around a pivot point: turn it right and the top rises as the bottom falls (brighter), turn it left for darker. It is the quickest way to change the overall balance of a track or a mix without thinking about bands.

High- and low-pass filters with selectable slopes clean up the extremes, and the tilt can work on the whole stereo signal, on the middle only or on the sides only. A response window shows the combined curve.

Reach for it for

- Quick brightness or warmth changes on a mix bus or master
- Matching the tonal balance of different takes or stems
- Brightening the centre or darkening the sides (MID / SIDE)
- Steep high- and low-pass clean-up
- Telephone and band-limited effects

Controls

Response window

Shows the combined curve of the filters, the tilt and OUTPUT from 20 Hz to 20 kHz, on a +/- 8 dB scale. The small triangle at the bottom marks the PIVOT.

High-pass

FREQ	The high-pass cutoff, 10 - 1000 Hz. Fully left = OFF.
Slope	The knob beside it: 6, 12 (default), 18, 24, 36 or 48 dB/oct. All slopes are Butterworth (maximally flat).

Tilt

TILT	The big knob. Right is BRIGHTER, left is DARKER; the word under it shows which, or FLAT at the centre. The value is the lift or cut at each end, up to 6 dB (12 dB end to end); hover over the knob to read it.
PIVOT	The balance point, 100 Hz - 5 kHz (800 Hz by default). Frequencies here stay where they are; the tilt rises on one side and falls on the other.
GENTLE / STEEP	GENTLE (default): a long, even 6 dB/oct slope across the whole range. STEEP: turns over more quickly around the pivot and leaves the far ends flatter.

Low-pass

FREQ	The low-pass cutoff, 1 - 22 kHz. Fully right = OFF.
Slope	The knob beside it: 6, 12 (default), 18, 24, 36 or 48 dB/oct, Butterworth.

Bottom row

STEREO / MID / SIDE	Where the tilt works: on both sides (STEREO, default), on the middle only or on the sides only - for example darken the sides and brighten the centre. The filters always work on both sides.
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TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	Click the TRIM knob to switch it; its LED sits at the knob's upper right. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The LED glows brighter while it is working.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter (3 ms latency, compensated by REAPER). CLIP is on by default and flashes red when it clips.
OUTPUT	Output level, -12 to +12 dB.
EQ IN	Switches the filters and tilt in or out (in by default). OUTPUT and the IN / OUT stages keep working.

The header

HORIZON / CLASSIC	Switches between the two looks: HORIZON (sky and sea, gold) and CLASSIC (cream and red). The sound is the same.
PRESETS	Opens the list of factory presets; the ◀ / ▶ keys step through them and the display shows the one in use (click it for the list too). Choosing one sets the tilt, filters, STEREO / MID / SIDE, OUTPUT and EQ IN; the input and output stages are left as they are. The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to

close it.

In REAPER's parameter list only

Bias / Sat low band / Sat high band / Ceiling

The rest of the output stage: the saturation's even-harmonic bias, where its low and high bands start (40 - 400 Hz and 2 - 16 kHz) and the clip ceiling or limiter threshold.

Presets

Mix Bus Lift

1.5 dB brighter around 1 kHz, with a gentle 24 dB/oct high-pass at 25 Hz.

Warm It Up

1.5 dB darker around 800 Hz, with a soft roll-off above 18 kHz.

Vocal Forward

3 dB brighter with the STEEP shape at 1.5 kHz, and an 18 dB/oct high-pass at 80 Hz.

Dark & Thick

3 dB darker at 600 Hz, a high-pass at 30 Hz and a low-pass at 9 kHz.

Bright Sides

2 dB brighter on the sides only, pivoting at 700 Hz.

Tame the Sides

2 dB darker on the sides only, STEEP at 2 kHz.

Telephone Band

No tilt; a band from 300 Hz to 3 kHz with 24 dB/oct filters.

Thin Out (steep HPF)

2.5 dB brighter at 400 Hz with a 48 dB/oct high-pass at 120 Hz.

Vintage Mid Roll-off

1 dB darker on the middle only, with a 24 dB/oct high-pass at 20 Hz and an 18 dB/oct low-pass at 14 kHz.

Clean Up (filters only)

No tilt; 36 dB/oct filters at 30 Hz and 20 kHz.

Reset all to defaults

Every control back to its starting value, including the input and output stages.

Tips

- Small moves go a long way: 1 - 2 dB of tilt is often enough on a full mix.
- Move the PIVOT to where the music's important energy sits; that part stays put while the balance shifts around it.
- Use EQ IN to compare, and adjust OUTPUT so the louder version doesn't fool your ears.
- SIDE with a darker tilt keeps wide cymbals and reverbs from getting harsh, without dulling the vocal in the centre.

File: gfx-tilt-eq.jsfx